Project Name: Injection Molding Acoustic Monitoring

Team Members:

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Problem (The "Why")
● **Current Issue:** Abnormal conditions in injection molding machines are difficult to detect early and often require manual inspection.
● **Impact:** Late detection can cause production disruptions, machine damage, and unexpected maintenance.

Proposed Solution
● **Our Approach:** Use two piezoelectric acoustic emission sensors to monitor acoustic signals from the injection molding machine and apply FFT to detect abnormal frequency patterns.
● **Alternative:** Use a microphone when AE sensors are unavailable, although accuracy may be lower due to environmental noise.
● **Key Value:** Provide low-cost, real-time machine condition monitoring through a web dashboard.

Core Features
● **Feature 1:** Real-time machine condition and FFT spectrum dashboard.
● **Feature 2:** Notification when an anomaly is detected.
● **Feature 3:** Monitoring history for previous measurements and anomaly events.

High-Level Architecture & Quality Assurance
● **Architecture Pattern:** Clean Architecture strictly adhering to SOLID principles.
● **Design Patterns Used:** Repository Pattern, Dependency Injection
● **Code Quality Goal:** Must Have 0 warnings as possible (scanned via static analysis), and 100% Unit Test Coverage.

Identified Challenges & Risks
● **Challenge:** A normal and damaged injection molding machine are needed as reference conditions to determine and validate the anomaly threshold.
● **Challenge:** Environmental noise can affect microphone measurements and reduce detection accuracy.
● **Challenge:** Changes in injection parameters or cycle conditions can alter the acoustic signal even when the machine is normal.
● **Risk:** Without verified damaged-condition data, the system may only be able to detect anomalies, not identify specific types of damage.